

CAP 6665 Project 1 Proposal

Document-to-Markdown Reconstruction

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This project will develop a document-to-Markdown reconstruction system that converts scanned or image-based document pages into structured Markdown for archival, indexing, search, and downstream text processing. The primary approach will use YOLO to detect semantic document regions such as titles, paragraphs, lists, tables, figures, and captions, followed by raw OCR to extract text from the detected regions. A spatial ordering module will then arrange the detected blocks into a likely reading order and apply Markdown formatting rules, converting titles into headings, text blocks into paragraphs, lists into bullet-style Markdown, tables into approximate Markdown tables when possible, and figures/captions into structured placeholders.

The project will use an open-source document layout dataset such as DocLayNet for training and evaluating the YOLO layout detector. The YOLO + OCR pipeline will be compared against a VLM-based approach that attempts to reconstruct Markdown directly from the page image. Evaluation will include layout detection performance using mAP and per-class precision/recall, OCR quality using character or word error rate on selected pages, and end-to-end Markdown reconstruction quality through manually inspected examples and structural similarity metrics such as correct ordering of headings, paragraphs, lists, tables, and captions.

Additional experiments will compare performance across different document types and degraded scan conditions such as blur, skew, compression, and low resolution. The expected result is a working prototype that evaluates whether a simple, interpretable YOLO + OCR pipeline can produce reliable structured documents for long-term preservation and retrieval, while also identifying which document elements are better handled by classical layout-aware processing versus direct multimodal generation.